

Student Academic Success, Subject Performance & Career Readiness Analytics Platform

AI-Powered Student Success Intelligence Platform

College / Department: FCAIT/MSc.IT (GLS University)

Team ID: 60

Team Members : Rahil Nagariya & Om Patel



The Problem: Student Data Exists, But Insights Are Fragmented

Institutions collect rich student data — yet datasets remain siloed, preventing integrated analysis.

Fragmented Data

Records stored across disconnected systems.

Hidden Learning Gaps

Subject-level weaknesses go undetected.

Difficult Early Risk Detection

No predictive models to flag at-risk students early.

Limited Career Guidance

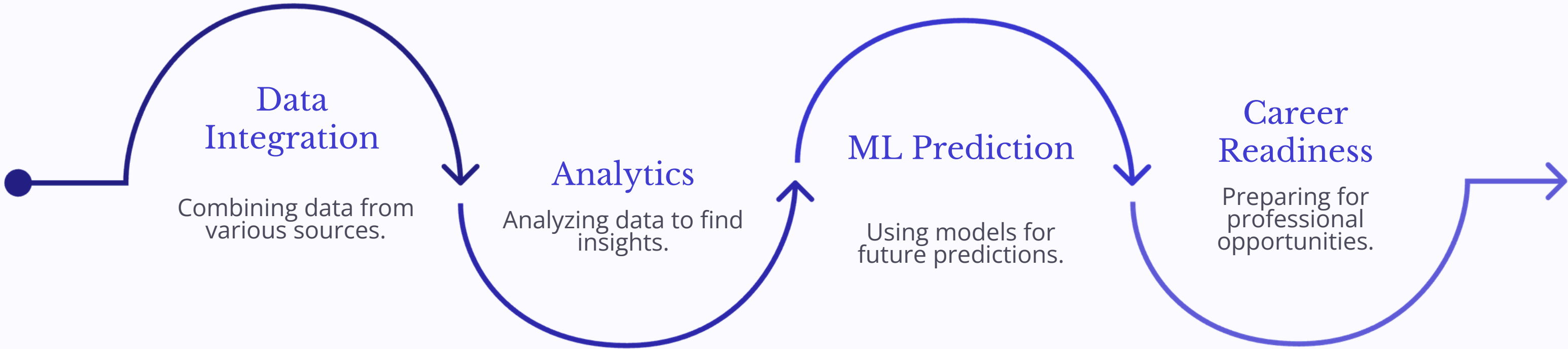
Career advice lacks data-driven personalization.

Historical Reporting Only

Descriptive reports replace predictive insights.

Our Solution: Student Success Intelligence Platform

An end-to-end platform with data stitching, warehouse design, ETL pipelines, ML prediction, GenAI insights, and Dockerized deployment.



Five core capabilities power the platform:

- 1

Data Stitching
Unify all datasets via a common Student ID.
- 2

Academic Analytics
Analyze performance trends and learning gaps.
- 3

Predictive Analytics
Predict outcomes and flag at-risk students.
- 4

Career Readiness
Calculate readiness scores and identify skill gaps.
- 5

GenAI Copilot
Generate personalized recommendations in plain language.

Also powered by: ETL Pipeline · Data Warehouse Design · Dockerized Deployment

End-to-End Technical Workflow



Data Flow Architecture

Raw institutional data flows through a structured pipeline — from ingestion and harmonization through parallel analytics and ML processing, GenAI interpretation, and finally role-based dashboards for every stakeholder.

Each stage is designed for modularity, allowing independent scaling and replacement of components as the platform evolves.

Technologies / Tools

A full-stack breakdown of the technologies powering each layer of the platform.

Layer	Technologies
Frontend	HTML, Tailwind CSS, Django Templates
Backend	Python, Django / FastAPI
Data Engineering	Pandas, NumPy
ETL & Data Stitching	Pandas, Python
Database	PostgreSQL
Data Warehouse	PostgreSQL, Star Schema / Analytical Schema
Data Analytics	Pandas, NumPy, SQL
Visualization	Plotly
Machine Learning	Scikit-learn, XGBoost
AI / GenAI Insights	Gemini API or OpenAI API
Deployment	Docker, Docker Compose
Version Control	Git, GitHub

From Fragmented Data to a Unified Student Profile

Proposed Star Schema

Student-related datasets are linked using Student_ID, while fact tables connect to relevant dimension keys.

Dimension Tables: Dim_Student, Dim_Subject, Dim_Department, Dim_Semester, Dim_Career, Dim_Skill, Dim_Time

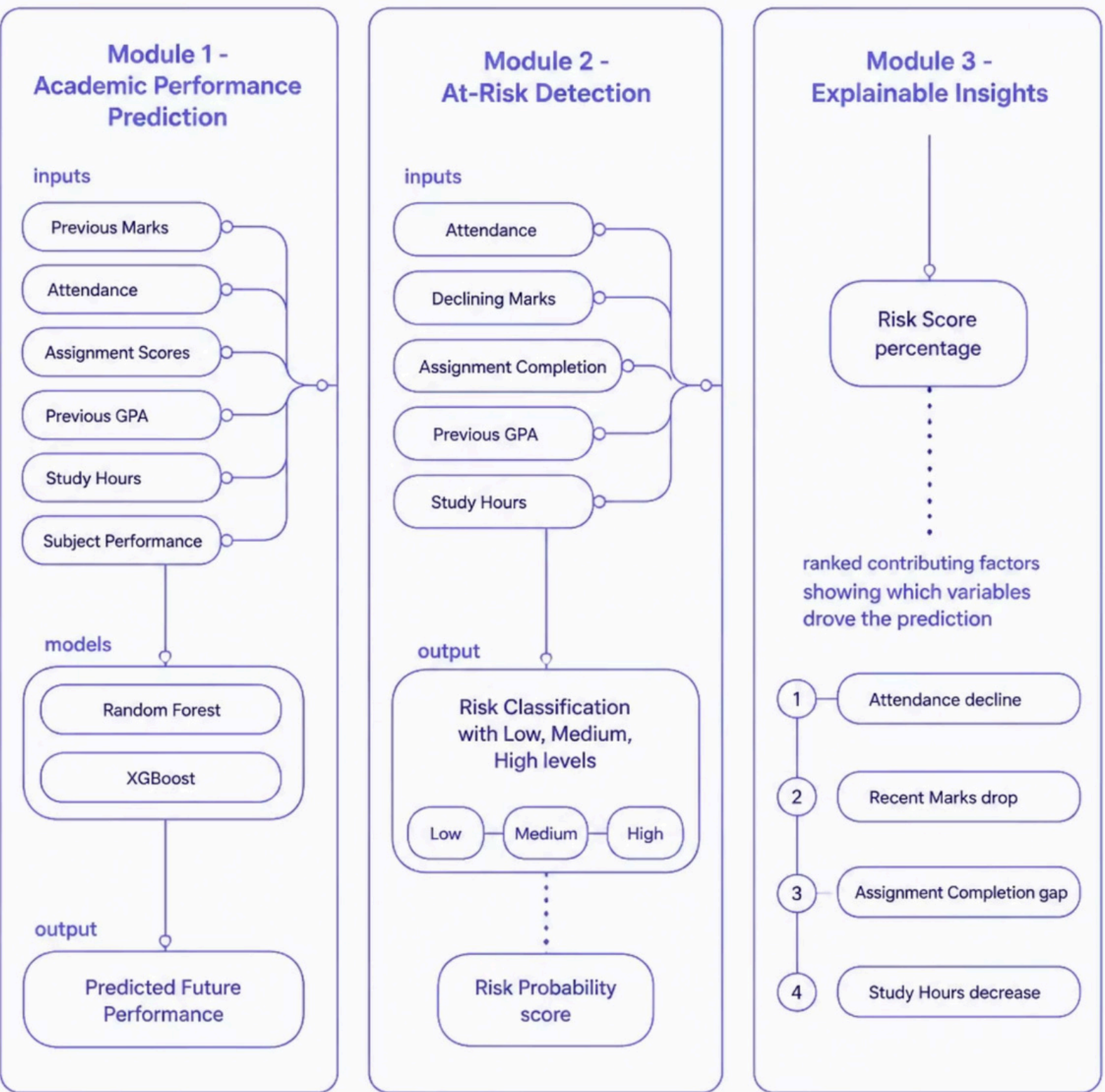
Fact Tables: Fact_Academic_Performance, Fact_Attendance, Fact_Career_Readiness

Example: Student S001

- Academic: 82%
- Attendance: 91%
- Study Hours: 3.5/day
- Python Skill: 4/5
- Target Career: Data Analyst

Data stitching converts isolated records into a **360° student profile** — enabling cross-domain analysis at the individual level.

Predictive Intelligence for Early Intervention



Prediction + Explanation

The proposed system delivers not just a risk score, but a transparent explanation of the factors driving each prediction — enabling faculty to understand why a student is at risk and what to do about it.

Example: Risk Probability: 87% — Top factors: Attendance ↓, Recent Marks ↓, Assignment Completion ↓, Study Hours ↓

From Prediction to Personalized Guidance

Career Readiness Score

76 / 100 · Target Career: Data Analyst

- Technical Skills — 30%
- Academic Performance — 20%
- Communication — 15%
- Problem Solving — 15%
- Projects — 10%
- Certifications — 5%
- Interview Skills — 5%

Identified Skill Gaps: Power BI, Statistics, Communication

GenAI Faculty Copilot

GenAI converts analytical results into understandable, actionable recommendations for every stakeholder.

Faculty

Which students need immediate intervention and why?

AI

12 students have high-risk scores. Major factors: low attendance, declining marks, incomplete assignments. Recommended action: schedule mentor meetings and provide targeted subject support.

Proposed Algorithm

Processing Pipeline

```
START
↓
Load multiple student datasets
↓
Validate and clean data
↓
Match records using Student_ID
↓
Transform and engineer features
↓
Load processed data into warehouse
```

ML & Output Loop

```
Train performance prediction model
Train at-risk classification model
Calculate career readiness score
↓
FOR each student:
    Predict performance
    Calculate risk probability
    Identify skill gaps
    Generate recommendations
↓
Display insights on dashboards
↓
END
```

📄 **Core Output per Student:** Prediction + Risk Level + Explanation + Recommended Action

Assumptions, Challenges & Expected Impact

Assumptions

- Unique Student ID per student
- Historical data available for training
- Academic and attendance data are reliable
- Career preferences provided by students
- External GenAI APIs are permitted

Challenges

- Missing and inconsistent data
- Data stitching across formats
- Class imbalance in at-risk detection
- Model accuracy and generalization
- GenAI reliability and grounding
- Student data privacy and security

Expected Impact

- Unified student data platform
- Early at-risk identification
- Subject-wise gap detection
- Personalized interventions
- Career readiness assessment
- Data-driven faculty decisions
- Actionable GenAI insights

Open Queries

- Data availability & access from the institution?
- GenAI API cost and rate limits at scale?
- Approval process for student data privacy?

Identify Problems Early → Understand Why → Recommend Action → Track Improvement

Thank You

Rahil Nagariya & Om Patel